Sri Ujwal Srinivas Varma Gunturi

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	9.08
Narayana Junior College	Class XII (Telangana State Board of Intermediate Education)	2023	98.10%
Jubilee Hills Public School	Class X (CBSE)	2021	96.33%

SKILLS

- **Programming Languages:** C, C++, Python, Bash Scripting, MySQL
- **Tools:** Linux, Git, Latex
- **Web Technologies and Database:** HTML, CSS, React.js, Node.js

WORK EXPERIENCE

Summer Internship at Samsung Research Institute Bangalore (SRIB) (2026)

End-to-End Egocentric Action Detection Pipeline:

- **Ego4D Dataset Filtering Pipeline:** Implemented a **self-correcting automated pipeline** to filter **false positives** from the Ego4D dataset and identify clips relevant to target categories.
- Designed a **confidence-aware verification loop** that reduced **manual dataset curation effort by 95%**.
- **Real-Time Action Detection:** Developed a **sliding-window action recognition pipeline** to predict **action labels in real time** from continuous video streams.
- Implemented **temporal smoothing** to reduce **prediction flickering** and produce smoother **action transitions** in real-time.

Teaching Assistant for the courses Probability and Random Variables, Artificial Intelligence, Operating Systems I and DBMS I

PROJECTS

Limit Order Book Matching Engine:

- Built a **C++17 multi-symbol Limit Order Book matching engine** implementing price-time priority, market orders, partial fills, cancellations, and exchange-style order modification semantics.
- Implemented **event-sourced persistence and crash recovery** with write-ahead command logging, deterministic replay

LOKUSQuant - A DSL for Quantitative Trading *(Course Work (CS3423), Prof. Ramakrishna Upadrasta)*

- Implemented a compiler pipeline with **lexical analysis, parsing, AST construction, hierarchical symbol tables, and semantic checks** including type, scope, and immutability validation.
- Developed **domain-specific semantic analysis** for trading strategies, supporting **technical indicators, market regimes, time-series operations, and trading signals**.

Document Management Repository *(Course Work(CS4443), Prof.Manish Singh):*

- Developed a **full-stack AI-powered document management system** with automated PDF metadata extraction, tagging, and semantic Vault-based document classification using **Groq LLaMA 3**.
- Implemented secure document storage and sharing with **AWS S3**, **JWT-based authentication, role-based access control**, storage quotas, and organization-based collaboration.

Multi-Client Quiz System with LLM Integration *(Course Work(CS3530), Prof. Praveen Tammana):*

- Developed a **multi-client TCP quiz system in C/C++** with multithreading, custom application protocol, real-time scoring, leaderboards, and keep-alive mechanisms.
- Integrated **LLM-based dynamic quiz generation** with server-side caching and performed **latency/throughput analysis using Wireshark packet traces** under varying client concurrency.

Memory Management System Calls in GemOS *(Course Work (CS3523), Prof. Saurabh Kumar):*

- Implemented system calls **(mmap, munmap, mprotect)** to efficiently manage virtual memory of a process
- Implemented **vm_area_pagefault** system call to allocate physical frames for a process using **lazy allocation**
- Extended munmap and mprotect to **update page table enteries** on unmapping and protection changes

RELEVANT COURSES

- **CSE Core Courses:** Deep Learning, Foundations of Machine Learning, Computer Networks, Software Engineering, Discrete Maths, Data Structures, Computer Architecture, Operating Systems (I & II), Algorithms, DBMS(I & II), Compilers(I & II), Artificial Intelligence
- **Math Courses:** Probability & Random Variables, Game Theory, Calculus I & II, Linear Algebra, Intro to Statistics

POSITIONS OF RESPONSIBILITY

- **Events Head**, Elan & nVision (Annual Techno and Cultural Fest) - 2025-26

Responsible for planning and organising vibrant events, including social impact initiatives, to enhance the fest experience.

- **Core Member**, Blockspace, Blockchain Technology Club of IITH - 2024-25

Explored the evolution of blockchain technology and actively promoted awareness and engagement within the campus.

- **Class Representative** for CSE'23 batch 2024-26

Acted as the liaison between faculty and students, facilitating communication, coordination, and academic support.

EXTRACURRICULAR

- **NSO** member of Athletics

- **NCC Cadet**, secured **A** grade in A-certificate examination, received appreciation trophy for exemplary performance